Event Study Result: Does Post-Earnings-Announcement Drift Show Up?

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> *This report tests whether post-earnings-announcement drift shows up in a pilot dataset of 261,537 U.S. equity earnings announcements (1996-01-02 to 2013-12-31), evaluated across 6 forward-return horizons (+1d, +5d, +10d, +20d, +40d and +60d) in both raw and market-adjusted form. A rank-normalized earnings-surprise measure -- available for the 236,174-event subset with a non-null surprise -- is used for the quintile sort and the per-event Spearman information coefficients that follow, at all 6 of those horizons.*

Data and methodology

The dataset. This report is computed on the event table at `data/events/equity_events_1996_2013.parquet` -- 268,606 U.S.-equity earnings announcements (1996-01-02 to 2013-12-31) built by joining the cleaned I/B/E/S earnings table (see `reports/earnings_data_report.pdf` for the surprise-construction detail) to CRSP daily-panel prices (see `reports/equity_data_audit.pdf` for the price-data detail). Of those announcements, 261,537 resolve to a real trading day 0 in the resolved permno's own trading history and form this report's population.

How "surprise" is defined. Every announcement can, in principle, be compared to the average Wall Street analyst estimate for that quarter (scaled by analyst disagreement -- the "analyst" measure) or to the same company's own result a quarter a year earlier (scaled by that company's own year-over-year volatility -- the mechanical "seasonal" measure, a fallback for announcements without usable analyst coverage). In this population, 182,712 announcements got an analyst-based measure, 53,462 fell back to the seasonal measure, and 25,363 had neither and carry no surprise value. The two measures are on different scales -- the analyst-benchmarked surprise has sample sd 9.008 and the seasonal-benchmarked surprise sd 3.363, a 2.68x dispersion gap -- so they are never pooled on their raw scales for anything scale-sensitive. Instead, both are converted to a percentile rank (0 to 1) separately within each (source, announcement calendar-quarter) group -- 144 such cells in total, median cell size 1,675 rows, smallest 9 -- and the pooled rank (`sue_rank_pct`) is what the quintile sort and the Spearman information coefficient below both use. This ranked subset covers 236,174 of the 261,537 day-0-ok events.

How PEAD is measured. "Post-earnings-announcement drift" is the hypothesis that stock prices keep drifting in the direction of a surprise for weeks after the announcement, rather than reacting all at once on day 0. This report tests it at 6 forward horizons -- +1d, +5d, +10d, +20d,

+40d and +60d -- each computed two ways: the raw forward return (the stock's own compounded return over the h trading days strictly after day 0) and the market-adjusted forward return (that same window's stock return minus the CRSP value-weighted market index's total-return compound over the same window). Day 0's own reaction-day return is deliberately excluded from every horizon, so nothing being reported here is the announcement-day pop; the whole point of measuring drift is to look AFTER the reaction day has already happened. Standard errors, t-stats and 95% confidence intervals are plain i.i.d. (`sd / sqrt(n)`, t-distribution, df = n-1) -- see the Limitations section below for what that inference does and does not support.

Point-in-time discipline. `sue_rank_pct` is ranked within each (source, announcement calendar-quarter) group, so no event's rank depends on the surprise distribution of a later period -- an event announced in, say, 2004Q1 only ranks against other 2004Q1 events from the same source, never against events from 2013 or any other later quarter. The forward-return computation is strictly forward from day 0 regardless of how the permno was resolved, and any missing daily return inside a window nulls that horizon rather than being silently treated as 0%. This is not a full look-ahead audit (see Limitations for what the remaining known simplifications are), but it does fix the specific look-ahead bug an earlier version of this rank had, where a 1996 event's quintile depended on 2013 data.

Results (does PEAD show up?)

This section reports the three market-adjusted checks that answer the question the whole report was set up to test -- whether PEAD shows up in this pilot dataset -- and its numbers are illustrated by the three charts that follow.

Per-horizon breakdown (6 horizons, all mkt-adj):
* At +1d: mkt-adj quintile sort is non-decreasing; Q5-Q1 spread is +0.671% (t = +22.84, significant at 5%); Spearman IC = +0.0661 (p < 0.05, p = 8.3e-226).
* At +5d: mkt-adj quintile sort is non-decreasing; Q5-Q1 spread is +1.182% (t = +20.64, significant at 5%); Spearman IC = +0.0712 (p < 0.05, p = 1.8e-261).
* At +10d: mkt-adj quintile sort is non-decreasing; Q5-Q1 spread is +1.462% (t = +18.84, significant at 5%); Spearman IC = +0.0675 (p < 0.05, p = 2.7e-235).
* At +20d: mkt-adj quintile sort is non-decreasing; Q5-Q1 spread is +1.755% (t = +16.60, significant at 5%); Spearman IC = +0.0608 (p < 0.05, p = 5e-191).
* At +40d: mkt-adj quintile sort is non-decreasing; Q5-Q1 spread is +2.242% (t = +15.23, significant at 5%); Spearman IC = +0.0566 (p < 0.05, p = 5.7e-165).
* At +60d: mkt-adj quintile sort is non-decreasing; Q5-Q1 spread is +2.905% (t = +14.86, significant at 5%); Spearman IC = +0.0567 (p < 0.05, p = 1.7e-164).

Read that block as three columns per row: (1) is the sort non-decreasing across Q1..Q5, i.e. does the mean return weakly rise with the surprise quintile? (2) is the Q5-Q1 spread large enough relative to its own standard error to reject a zero-spread null at 5%? (3) is the pooled per-event Spearman rank correlation between surprise rank and forward return large enough relative to its own sample size to reject a zero-correlation null at 5%? All three must line up for a

horizon to fully corroborate a positive-PEAD reading; any single failing check leaves that horizon partially supported at best.

Conclusion. At every one of the 6 horizons tested, the mkt-adj quintile sort is non-decreasing, the Q5-Q1 spread is significantly positive at 5%, and the pooled Spearman IC is significantly positive at 5%. The data supports a positive PEAD monotonicity signal at every horizon in this population.

Limitations (what this result does not support)

This section names, in one place, everything this pilot's numbers do NOT support. Every point below traces back to a specific decision or boundary in the upstream data, not a flaw in the statistical computation itself; the point of naming them here is to keep any "headline" reading of the numbers honest about their scope.

Standard errors are plain i.i.d. Every t-stat, standard error and confidence interval reported here uses `sd / sqrt(n)`, not clustering by firm or by announcement date, even though earnings cluster heavily in fiscal-quarter "seasons." To the extent nearby events are correlated (same firm, same season, same industry shock, ...) these standard errors understate the true uncertainty; a later, more rigorous phase (see `memory/phase1-full-equity-pead-plan.md`) handles clustered/bootstrapped inference. These numbers should be read as a first look, not a final significance verdict.

No delisting-return handling. When a stock is delisted (going private, acquired, or dropped by the exchange) mid-horizon, its own price series simply ends -- CRSP's `dsedelist` delisting-return series is not merged into the return panel this report is built on. The longer-horizon numbers implicitly condition on survival to the horizon end. Whether attrited events differ systematically from retained ones in their earnings-surprise distribution is not audited in this report; the conservative reading is that this conditioning is not guaranteed random.

No transaction-cost or turnover analysis. This report measures whether a drift statistically exists on paper. It does NOT price out the commissions, spreads, and turnover cost of trading on it, and does not simulate any actual portfolio construction. A "large" Q5-Q1 spread here does not translate into an implementable strategy -- that is the entire scope of Phase 2 (see `memory/phase2-equity-strategy-plan.md`), not this report.

Pilot universe scope. The population is the resolved event universe defined and enumerated in the upstream reports (`reports/equity_data_audit.pdf` and `reports/earnings_data_report.pdf`) -- not the CRSP universe, not the full I/B/E/S coverage, and not any exchange-index index membership at any point in time. Any inference from these numbers to a specific listed-equity universe (large-cap, mid-cap, index-member, etc.) requires cross-checking against the upstream reports' own scope documentation.

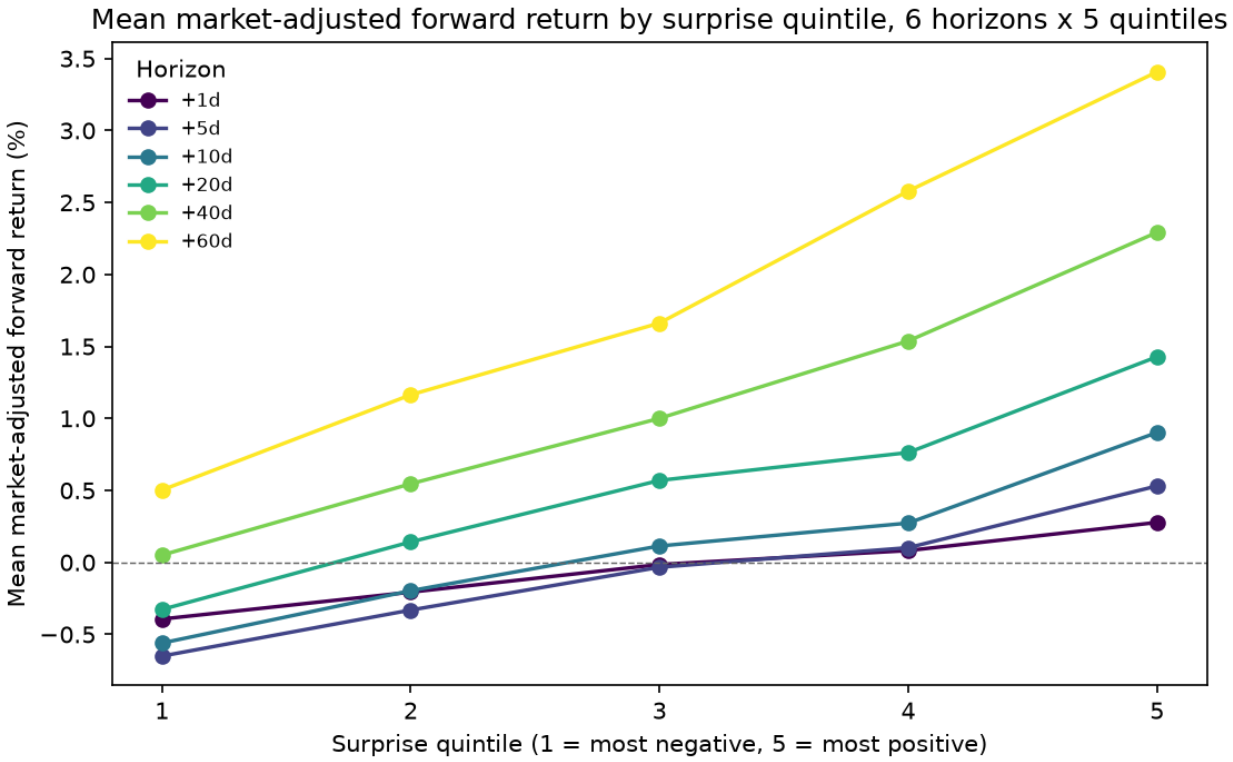
Quintile-mean Spearman rho is descriptive, not inferential. The quintile-index-vs-quintile-mean-return Spearman rho reported alongside the quintile sort (in sub-slice 10b) is computed over only 5 quintile means and, more fundamentally, those 5 means are not 5 independent draws -- so no p-value is reported next to it, and its value should be read as a compact summary of

monotonicity, not a significance test. The real inferential test for a monotone relationship in this report is the pooled per-event Spearman information coefficient (also in sub-slice 10b), which uses the full ranked-event sample.

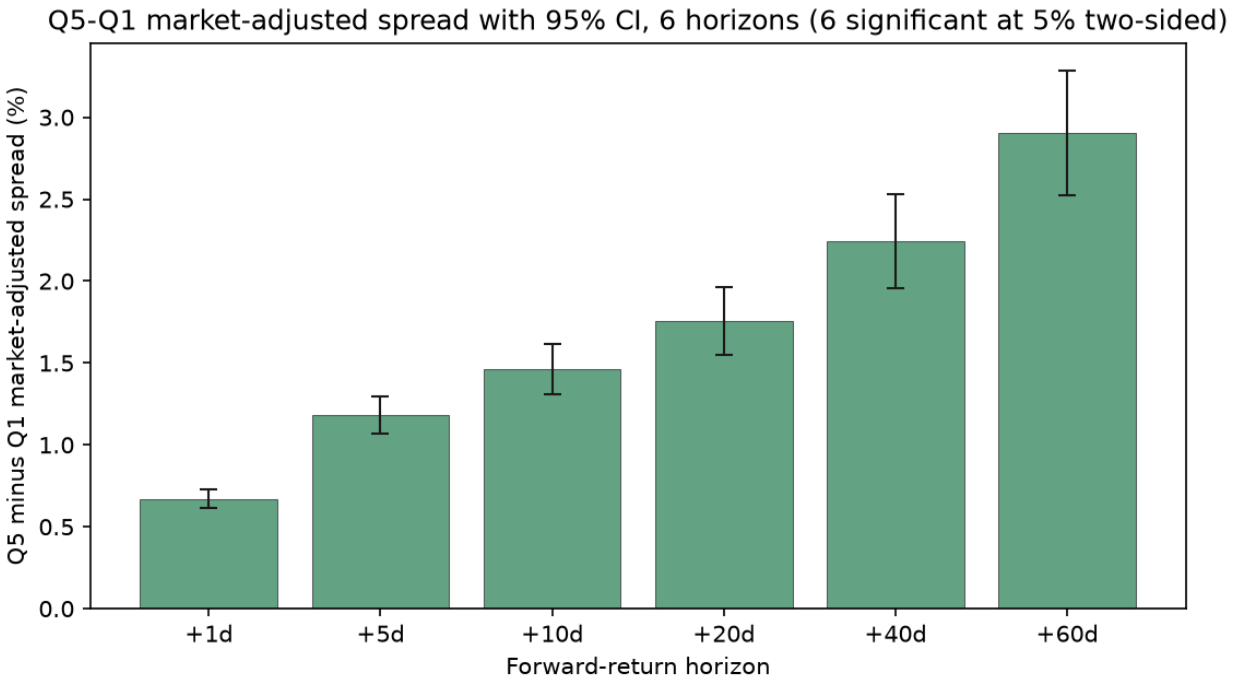
Point-in-time is fixed for the specific bug it fixed, not everything. The ranked surprise (`sue_rank_pct`) is grouped by announcement calendar quarter, so a 1996 event's rank no longer depends on 2013 surprise data -- that specific look-ahead bug is gone. But ranking still uses the whole calendar quarter's cross-section (up to ~3 months of same-quarter data, including events announced after the event being ranked), which is the standard PEAD construction, not a strictly real-time-as-of-announcement cutoff. This report does not claim to have audited every other potential look-ahead path through the upstream data.

The per-horizon summary-statistics table (in the Results section, sub-slice 10b) runs on the full 261,537-event day-0-ok population, while the quintile sort and the Spearman IC further below run on the 236,174-event subset with a non-null surprise -- a difference of 25,363 events, comprising the 25,363 announcements with neither an analyst-based nor a seasonal-based surprise value. Any comparison of "the mean at horizon h" from one table with "the Q5-Q1 spread at horizon h" from the other implicitly conditions on that missing subset.

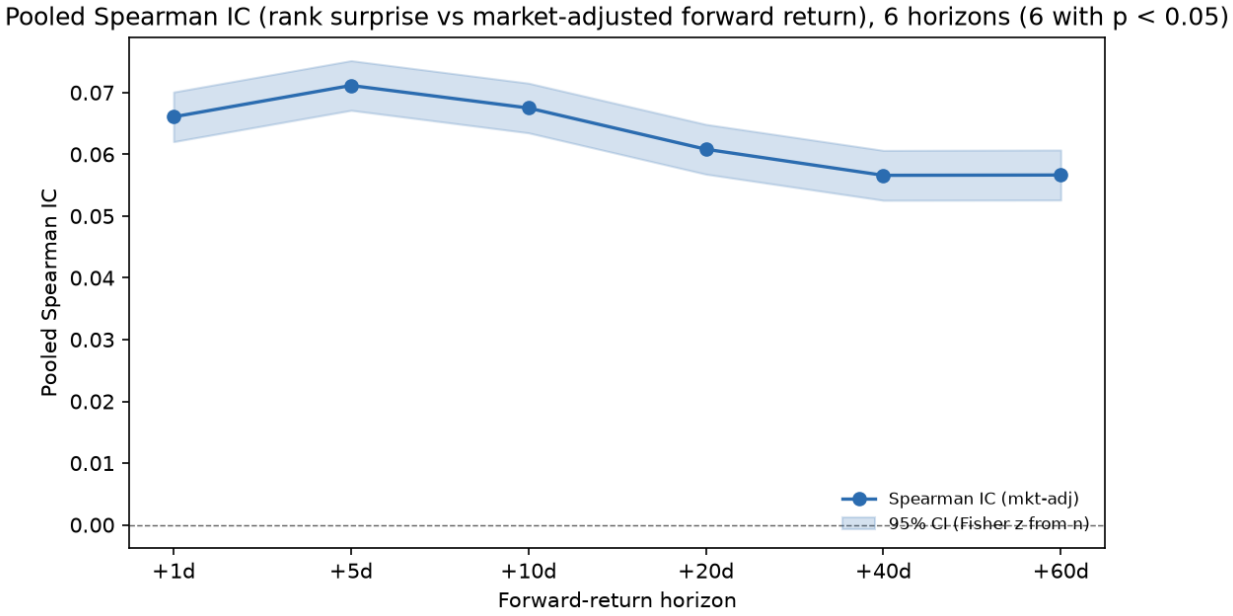
Non-independence: 6,797 of the 261,537 events used here (2.6%) share a (permno, day-0-date) pair with at least one other event, and 1,151 (0.4%) carry a permno also claimed by a different I/B/E/S ticker over an overlapping window -- neither is deduped or filtered here, matching the upstream event table's tag-don't-filter convention, so a handful of observations are not strictly independent of each other, which the i.i.d. standard errors above do not account for.



Each line is one forward-return horizon; the x-axis is the surprise quintile (1 = most negative, 5 = most positive) and the y-axis is the mean market-adjusted forward return in percent. The mkt-adj quintile sort is non-decreasing from Q1 to Q5 at all 6 horizons. At the longest horizon shown (60d), Q1's mean is +0.50% and Q5's mean is +3.41%; the sign, ordering and inter-line spacing here are the numbers the Q5-Q1 spread and Spearman IC charts below quantify with confidence intervals.



Each bar is the market-adjusted mean return in the top surprise quintile minus the mean in the bottom, at that horizon; error bars are the +/-1.96 standard-error 95% CI. The spread widens with horizon (from +0.67% at 1d to +2.90% at 60d), spans +0.67% at 1d to +2.90% at 60d, and clears the 5% two-sided significance bar (|t| > 1.96) at all 6 horizons. Green marks a positive spread, red a negative one; the sign is picked from the data, not fixed.



Solid line: the pooled Spearman rank correlation between per-event surprise rank (`sue_rank_pct`) and per-event market-adjusted forward return, computed on the full ranked subset at each horizon; shaded band: 95% CI derived from a Fisher z-transform of the sample size (Slice 9's JSON reports IC and p-value but not an explicit CI). IC ranges from +0.057 at 40d to +0.071 at 5d; the magnitude shrinks from |IC|=0.066 at 1d to 0.057 at 60d. The 5% two-sided significance bar (p < 0.05) is cleared at all 6 horizons.